Do Now · retrieval
A*: heuristic search I

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Why are predecessors recorded when a distance improves?

Q2

Can Dijkstra assume negative edge weights are safe?

Q3

What is 2+3+2?

Q4

Starting S=0, edge S–A=4, then A–G=2. What is the S→A→G path cost?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

g, h and f have different meanings — g is distance already travelled; h estimates the remaining distance; f=g+h is the estimated total.

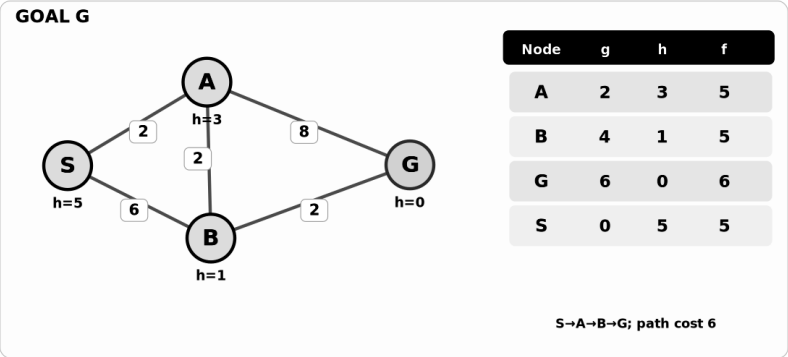
Core idea 2

Start with a frontier — From S, A gets g2,h3,f5, while B gets g6,h1,f7. Choose A, not simply the smallest h.

Core idea 3

Improve a discovered route — From A, B improves to g4,h1,f5. From B, G improves to g6,h0,f6.

Key visual · complete concept



New Learning · applying the ideas

Core idea 4

Stop on the selected goal — When G is selected from the frontier, reconstruct S→A→B→G with total cost 6.

Core idea 5

Keep the heuristic safe — The estimates h(S)=5,h(A)=3,h(B)=1,h(G)=0 never exceed the actual cheapest remaining cost.

Vocabulary · use the correct technical terms

g(n) — Known best cost from the start to vertex n.

h(n) — Estimated cost from vertex n to the goal.

f(n) — Combined score g(n)+h(n).

frontier / open set — Discovered vertices awaiting expansion.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

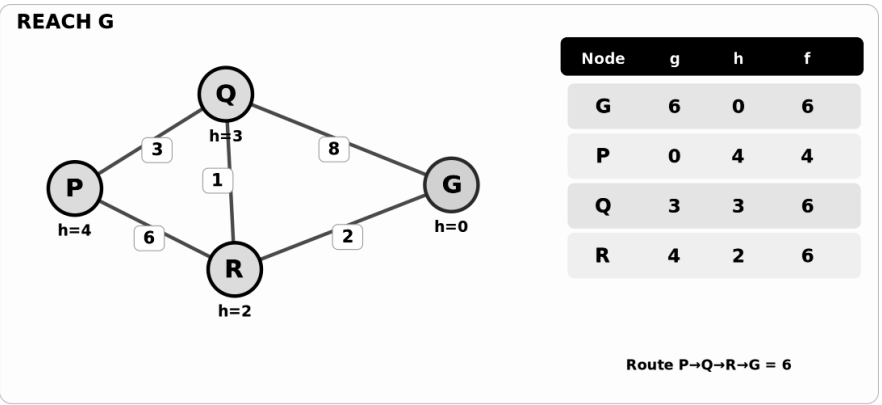
Problem

Trace A* on P, Q, R, G using the supplied admissible estimates.

Follow the worked method

1. Predict the next state — From P, Q has g3+h3=f6, R has g6+h2=f8. Choose Q.
2. Update the table — From R, G improves to g6,h0,f6.
3. Update the table — Select G: predecessor chain yields P→Q→R→G, cost 6.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict the next state
From M, choose N because f(N)=2+3=5 < f(O)=5+1=6.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

A node has g=7 and h=3. Give f and define each value.

Question 2

Graph edges S–A=2, S–B=5, A–B=1 and B–G=2. Heuristics: h(S)=4,h(A)=2,h(B)=1,h(G)=0. Calculate f for A and B when first found and state the optimal S→G route.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why admissibility matters. For the graph S–A=2, A–G=4 and S–G=10, what is the greatest admissible h(A), and why?

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A* chooses the next vertex solely by its smallest heuristic h(n).

Use key vocabulary from memory

1. Explain g(n) in your own words.

2. Explain h(n) in your own words.

3. Explain f(n) in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.